

EV Adoption Around the World

Modelling with the S-Curve

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Climate Change Calculated / Klimawandel Nachgerechnet
A channel for educating, calculating, and modeling anything climate change related.

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How can we calculate who's next?

EV-Mobility-Modelling

A Comparative Study of S-Curve Modelling of EV Adoption and the Role of National Policies

Climate Change Calculated, TU Ilmenau

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Motivation: Why is this debate happening now?

Why is the European automobile industry trying to water down the combustion-engine phase-out in 2035,
even though it is still about 10 years in the future —
and countries like Norway have almost already reached this goal?

Motivation: Why is this debate happening now?

To answer this, we need to address three questions:

- How can we predict and estimate the coming years?
- How do we judge the current situation?
- Which signals matter — and which are just noise?

We need a method that is simple, transparent, and based on data — not wishful thinking.

The solution: Modeling technology adoption

A powerful way to understand technological change is to look at past transitions.

From horses to cars, from analog to digital photography, from simple mobile phones to smartphones — the adoption follows a characteristic S-shaped curve.

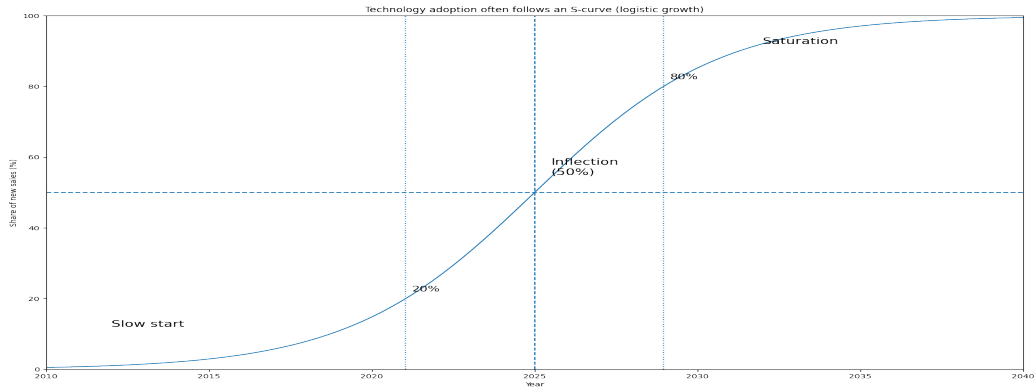
The S-curve is a surprisingly powerful yet simple model, with only a few parameters:

- the growth rate
- the final saturation level
- and the tipping point in time

Policies can influence all of these parameters — but they cannot change the basic S-shape itself.

Most technologies grow like an S

- Slow start → inflection (tipping point) → saturation.
- Shares are better than raw counts for comparing countries.



The logistic model

$$s(t) = \frac{L}{1 + e^{-k(t-t_0)}}$$

- L : saturation (max share).
- k : growth speed.
- t_0 : inflection year (50%).

We just need to determine these parameters such that the curve fits the observed data points.

Fitting the logistic model

The S-curve can

- be early or late, with t_0
- have slow or fast growth, with k
- have less than 100% saturation, with L

Open data (reproducible)

- Global: IEA Global EV Data Explorer
- Hannah Ritchie (2024) - "Tracking global data on electric vehicles" Published online at OurWorldinData.org. Retrieved from: <https://archive.ourworldindata.org/20251208-170037/electric-car-sales.html> [Online Resource] (archived on December 8, 2025).
- More sources in the description and in: <https://github.com/TUIIlmenauAMS/ClimateChangeCalculated>

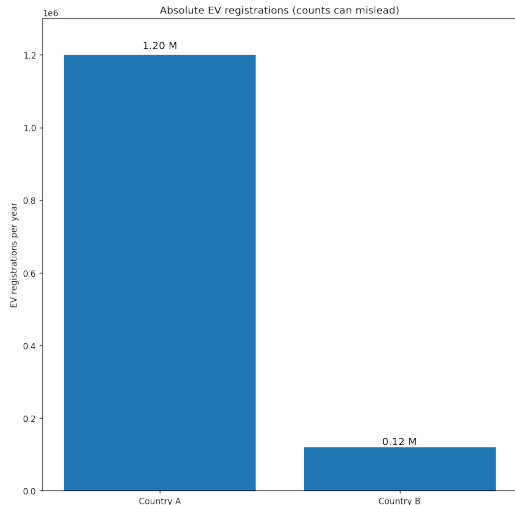
Open data (reproducible)

- All plots are generated from CSVs and code linked in the description.
- Code was written with the help of our "**Climate Change Assistant**".
- This assistant can also be used at any time for background questions <https://chatgpt.com/g/g-6829ff3e69cc81918c917c75c6925b35-climate-change-assistant>



What counts as an EV?

- BEV: Battery Electric Vehicle
- PHEV: Plug-in Hybrid
- Main model: BEV+PHEV share of new registrations

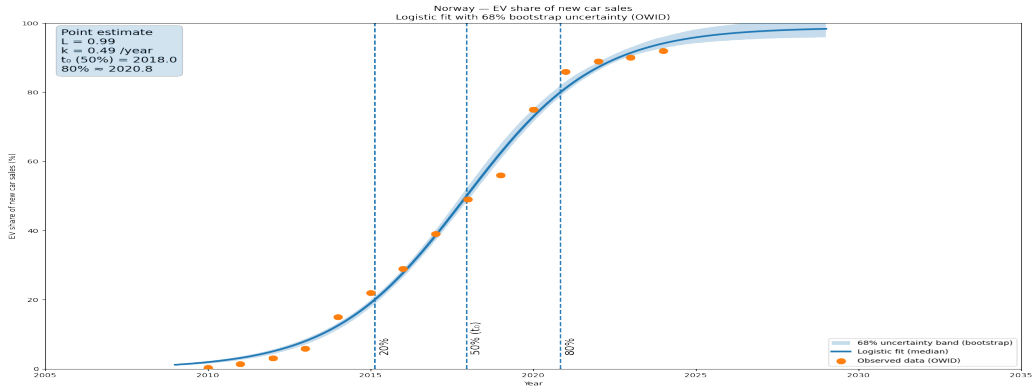


Fitting S-curves (what we do in Python)

- Compute BEV+PHEV share = $\frac{\text{BEV+PHEV regs}}{\text{total regs}}$.
- Fit logistic curve with constraints: $0.5 \leq L \leq 1.05$, $k > 0$.
- Solve milestone years for 20%, 50%, 80%.
- **Bootstrap resampling** to get 68% uncertainty bands ($\pm$ standard deviation).
- The shaded plot area shows the **uncertainty bands** of the fitted S-curve.

Example: Norway (late stage)

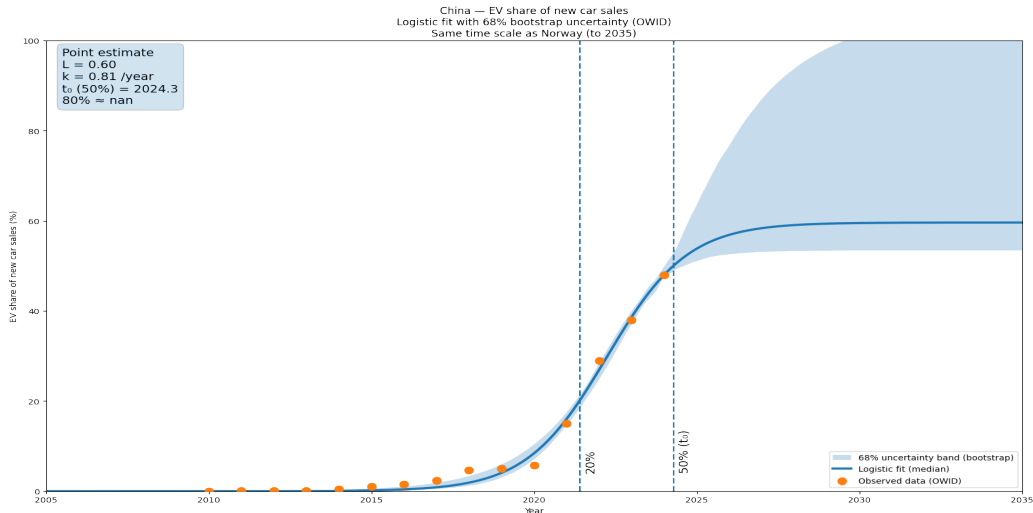
Near saturation: the curve must flatten. The shaded area is the uncertainty band.



Show milestones: t_{20} , t_{50} , t_{80} (median + 68% CI).

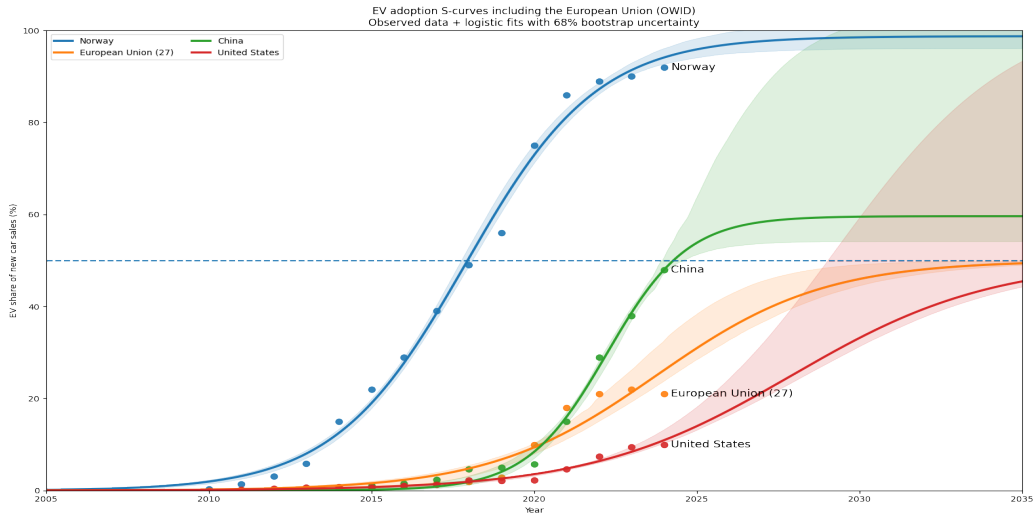
Example: China (mid stage)

Huge market, strong mid-curve acceleration. More uncertainty towards the future.



Multi-country comparison

Different positions on the same S-pattern. The EU converges to low saturation.



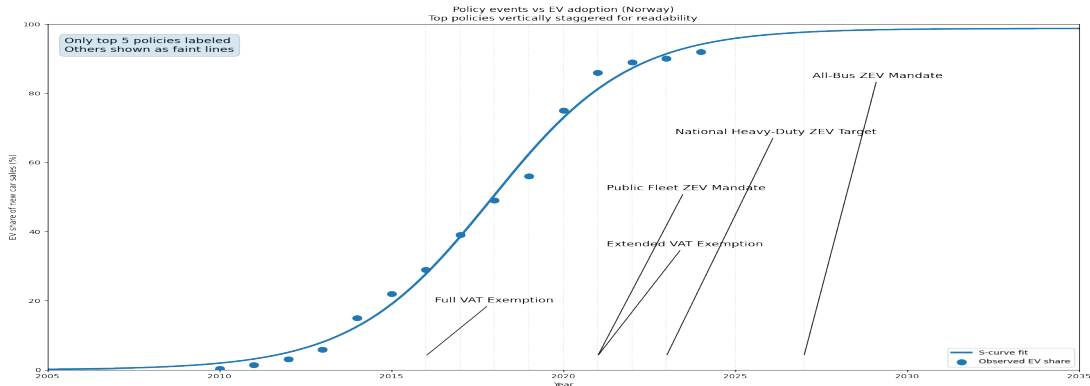
The 2035 Goal of 100% EV

Observe:

- Norway as almost there.
- China has a good chance of reaching this goal.
- Even the US has a realistic chance of reaching this goal, based on current data,
- but Europe appears to be approaching 50% saturation, which means it will be difficult to achieve this goal, and new **policy measures** will be necessary.
- Hence the attempt by the European automotive industry to water down the target.

How policy interacts with the EV S-curve: Norway

Policy does not create adoption — it shifts and accelerates it.



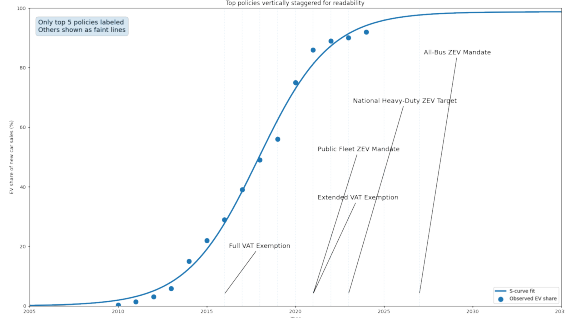
Only the most important policy interventions are labeled; others are shown as faint lines.

Policy and the EV S-curve: Norway vs United States

Norway applied strong incentives much earlier than the US.

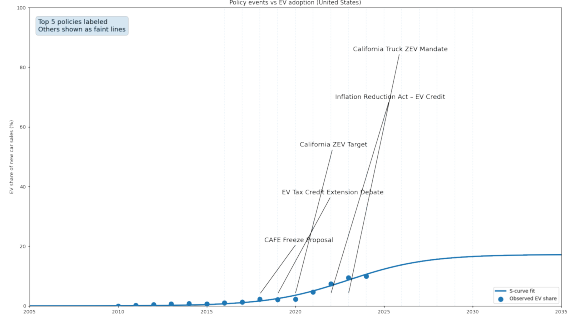
Norway

Policy events vs EV adoption (Norway)
Top policies vertically staggered for readability



United States

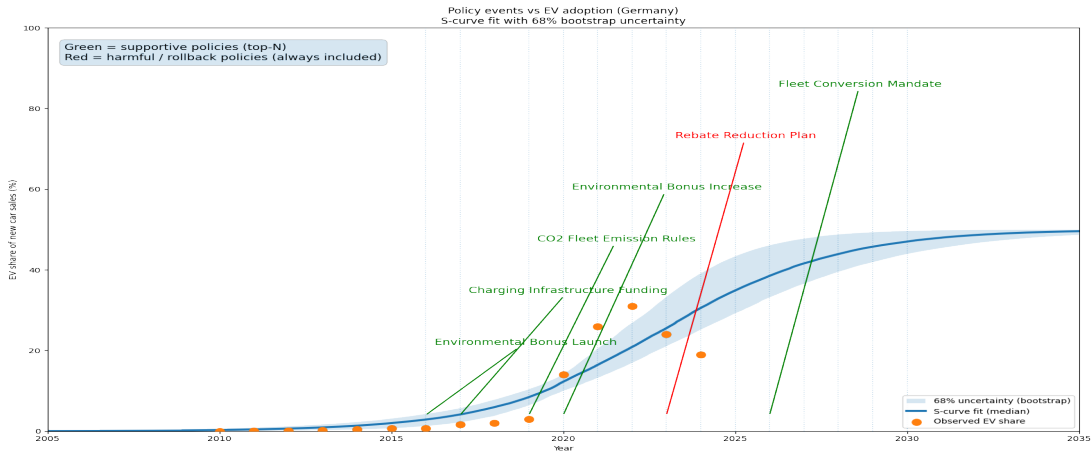
Policy events vs EV adoption (United States)



Only the 3–5 most important policy events are labeled; others are shown as faint vertical lines.

Germany: Policy events and EV adoption uncertainty

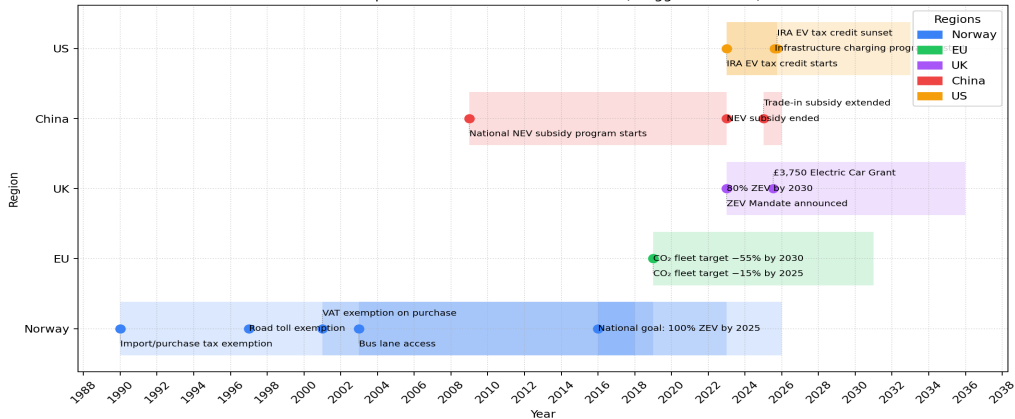
Germany is still early on the S-curve, hence policy consistency is important this stage.



International Policy Timelines

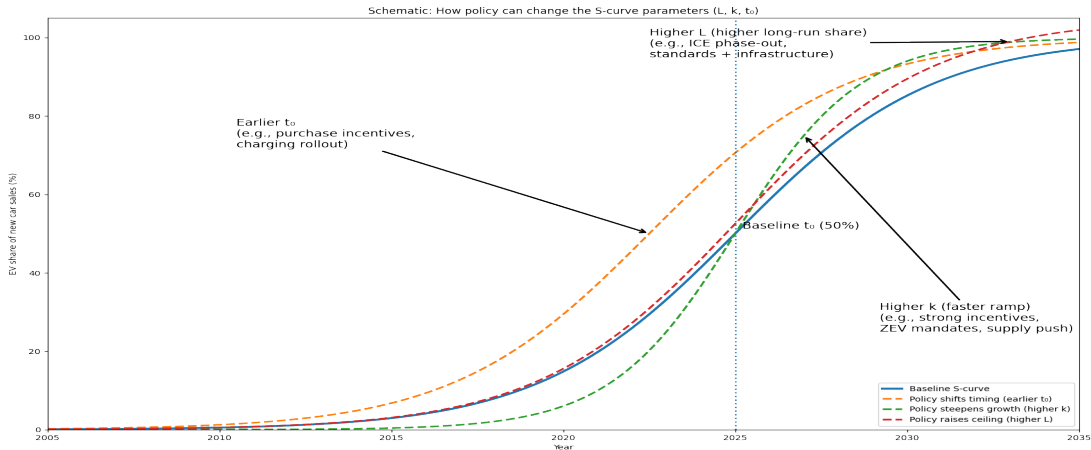
- Policy shocks can change k quickly.
- Supply constraints and charging rollout matter.

EV Adoption Policies with Timeline Bands (Staggered Labels)



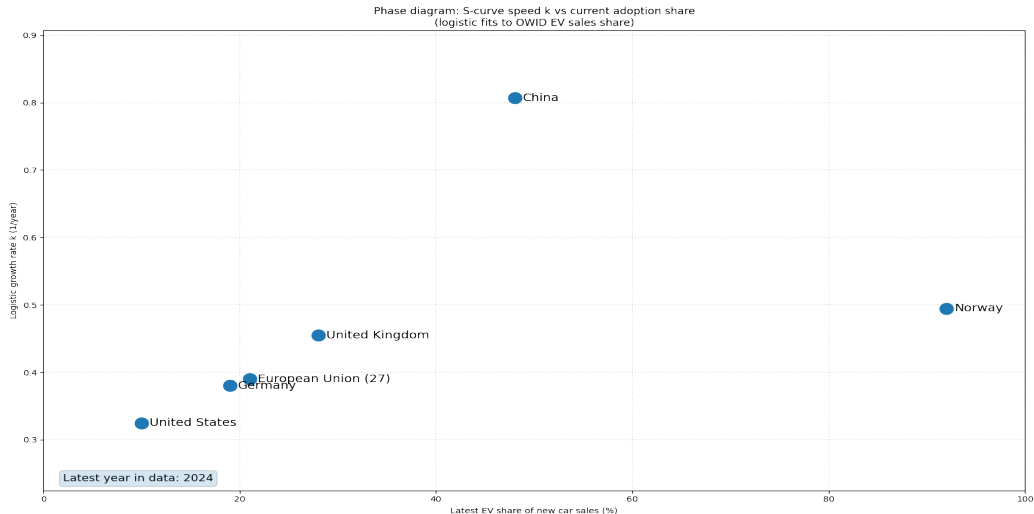
Schematic: How policy changes the S-curve (L , k , t_0)

Policy can shift timing (t_0), steepen growth (k), or raise the ceiling (L).



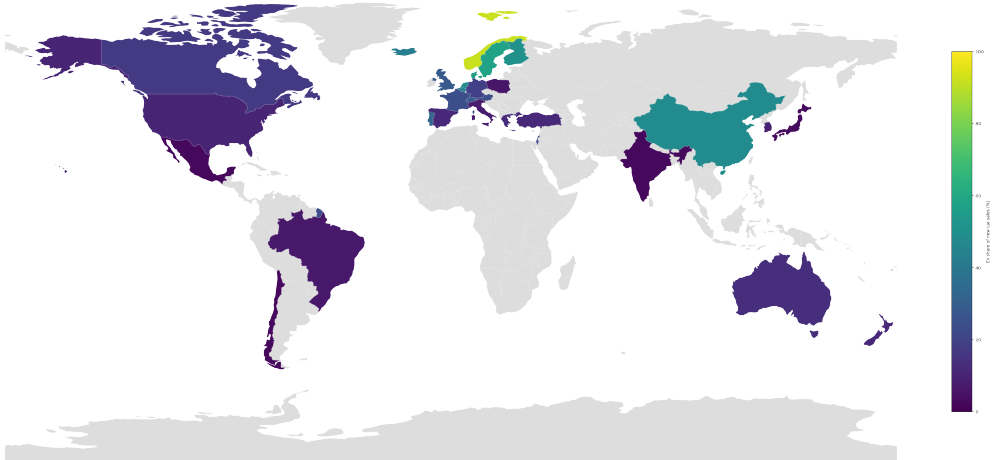
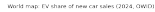
Phase diagram: level vs speed

Who is early-but-fast vs late-and-slow?



World map: current BEV share

Where is adoption already high?



Countries with data: 30 • Grey = no data

Takeaways

- Many countries are entering the steep part of the S-curve.
- Falling EV costs and better charging push adoption faster.
- Uncertainty is explicit: we show confidence bands.

Data beats opinions.

- The S-curve allows us to estimate the current development dynamics and extrapolate them into the future.
- Political measures influence the start, the speed of the dynamics, and the saturation at the end.
- The EU's current S-curve indicates insufficient saturation, suggesting that supportive policy measures are needed (purchase incentives, legal concessions, combustion engine phase-out, infrastructure, etc.).

Call to action

Try your own country: code + data in the description.

- Comment: who do you think is next?
- Subscribe for more reproducible calculations.